

Verification Report: VHDL FFT Core

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This report provides a technical snapshot of the current state of the **Radix-2 FFT/IFFT processor**. The design utilizes a **Block Floating Point (BFP)** architecture to maintain high dynamic range while using 16-bit fixed-point arithmetic. This automated verification suite ensures that the mathematical integrity of the core remains intact across various input scenarios and transform lengths.

Core Metrics Explained

- **SQNR (Signal-to-Quantization-Noise Ratio):** Quantifies the ratio between the signal power and the noise introduced by rounding and quantization. Higher values indicate lower mathematical degradation.
- **ENOB (Effective Number of Bits):** Translates the SQNR into an equivalent bit-depth. A result of **14.66 bits** represents the theoretical maximum for this 16-bit architecture when capped at a 90 dB floor.
- **Block Exponent:** Tracks the number of arithmetic shifts performed by the BFP logic to prevent overflow. This metric validates that the hardware is actively scaling data to preserve headroom.
- **Quantization Noise Floor:** Small-scale signals (e.g., -40 dB) naturally exhibit lower SQNR/ENOB. This is an expected result of fixed-point arithmetic, demonstrating the noise floor when only a fraction of the 16-bit dynamic range is utilized.

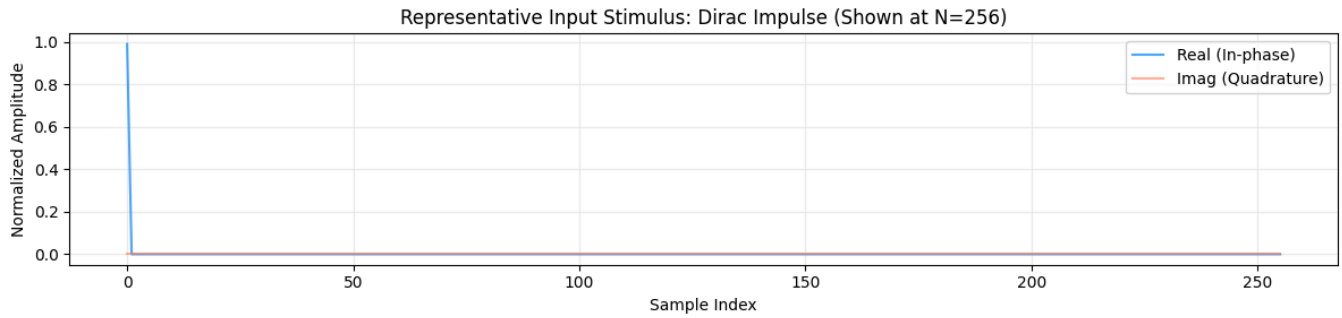
Report Structure

The following sections provide a detailed visual analysis of the core's performance, including clock cycle efficiency and spectral purity for various test vectors. The report concludes with a **Final Summary Table**, which aggregates the results of the entire regression suite—including a system-wide FFT/IFFT round-trip—to provide a definitive pass/fail status for the current hardware build.

Verification Summary

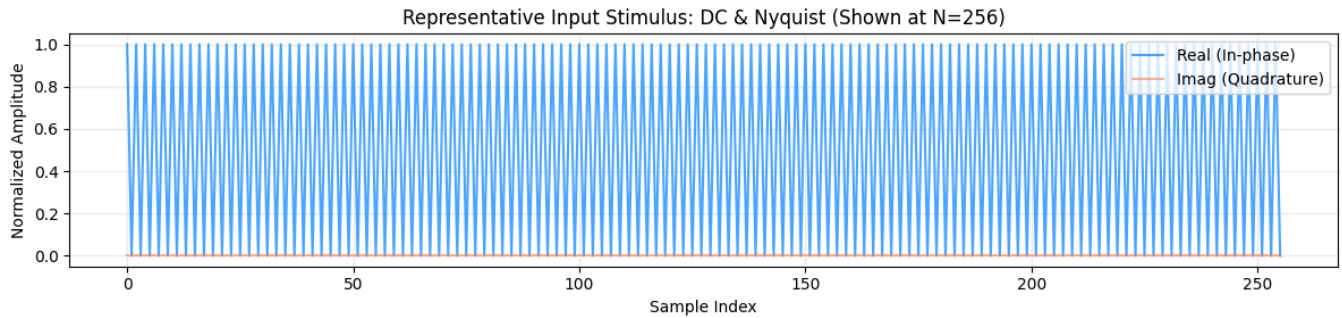
The regression suite evaluates the core against impulses, white noise, and small-signal sensitivity tests ("Torture Tests") for sizes ranging from $N = 16$ to $N = 2048$. Stable performance across these scenarios confirms that the implementation is mathematically verified and ready for hardware adaptation.

--- Scenario: Dirac Impulse ---



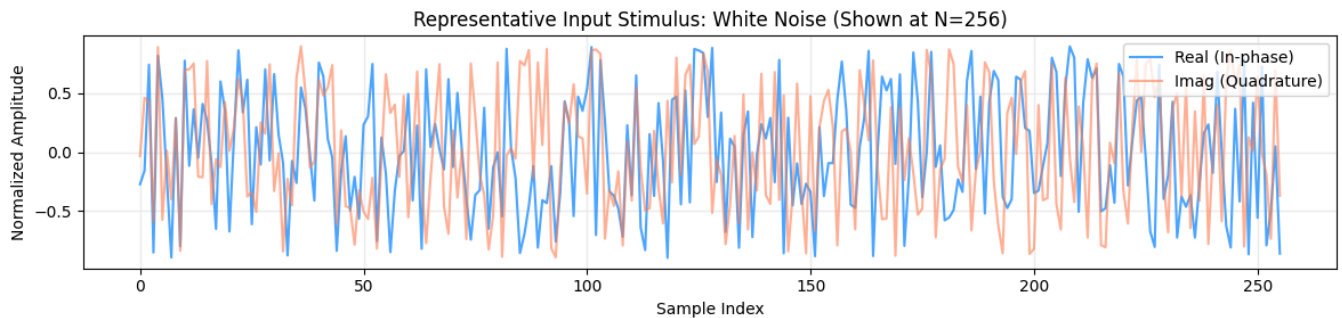
✓	N=16		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=32		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=64		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=128		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=256		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=512		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=1024		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66
✓	N=2048		SQNR: 90.00 dB		Max Err: 9.77e-06		Exp: 1		Enob: 14.66

--- Scenario: DC & Nyquist ---



✓	N=16		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 4		Enob: 14.66
✓	N=32		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 5		Enob: 14.66
✓	N=64		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 6		Enob: 14.66
✓	N=128		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 7		Enob: 14.66
✓	N=256		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 8		Enob: 14.66
✓	N=512		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 9		Enob: 14.66
✓	N=1024		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 10		Enob: 14.66
✓	N=2048		SQNR: 90.00 dB		Max Err: 0.00e+00		Exp: 11		Enob: 14.66

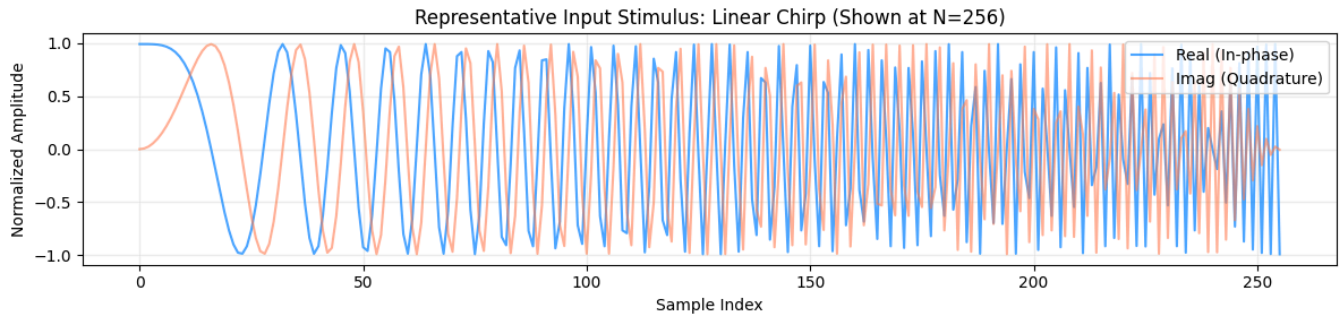
--- Scenario: White Noise ---



✓	N=16		SQNR: 72.97 dB		Max Err: 1.13e-03		Exp: 4		Enob: 11.83
✓	N=32		SQNR: 72.54 dB		Max Err: 1.74e-03		Exp: 4		Enob: 11.76
✓	N=64		SQNR: 70.95 dB		Max Err: 3.17e-03		Exp: 5		Enob: 11.49
✓	N=128		SQNR: 67.10 dB		Max Err: 9.02e-03		Exp: 6		Enob: 10.85
✓	N=256		SQNR: 67.76 dB		Max Err: 1.29e-02		Exp: 6		Enob: 10.96

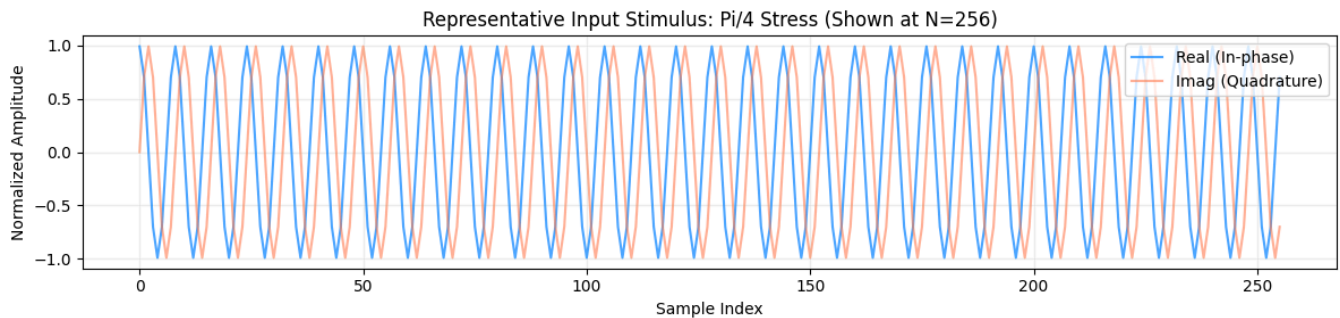
✓	N=512		SQNR: 66.33 dB		Max Err: 2.15e-02		Exp: 7		Enob: 10.73
✓	N=1024		SQNR: 65.07 dB		Max Err: 4.41e-02		Exp: 7		Enob: 10.52
✓	N=2048		SQNR: 64.43 dB		Max Err: 7.09e-02		Exp: 8		Enob: 10.41

--- Scenario: Linear Chirp ---



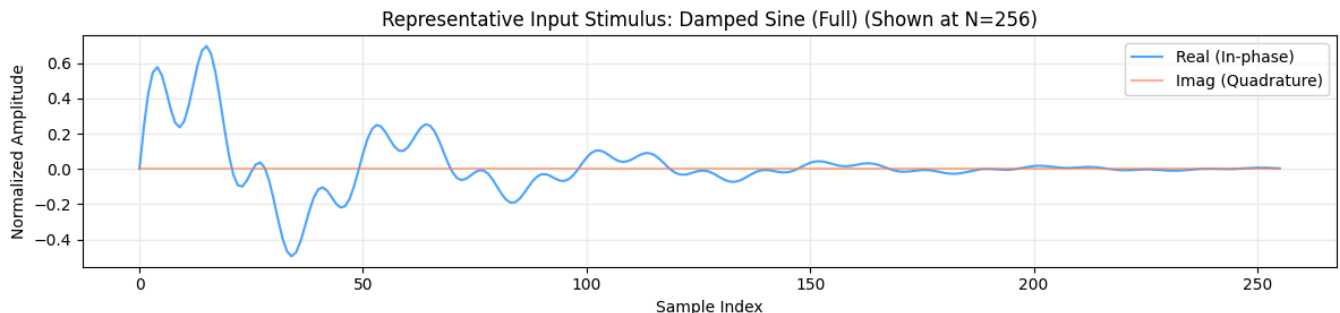
✓	N=16		SQNR: 76.18 dB		Max Err: 8.34e-04		Exp: 4		Enob: 12.36
✓	N=32		SQNR: 77.25 dB		Max Err: 1.27e-03		Exp: 4		Enob: 12.54
✓	N=64		SQNR: 74.87 dB		Max Err: 3.57e-03		Exp: 5		Enob: 12.15
✓	N=128		SQNR: 74.48 dB		Max Err: 6.37e-03		Exp: 5		Enob: 12.08
✓	N=256		SQNR: 73.43 dB		Max Err: 8.32e-03		Exp: 6		Enob: 11.90
✓	N=512		SQNR: 71.01 dB		Max Err: 1.35e-02		Exp: 6		Enob: 11.50
✓	N=1024		SQNR: 70.73 dB		Max Err: 3.04e-02		Exp: 7		Enob: 11.46
✓	N=2048		SQNR: 68.09 dB		Max Err: 5.40e-02		Exp: 7		Enob: 11.02

--- Scenario: Pi/4 Stress ---



✓	N=16		SQNR: 90.00 dB		Max Err: 1.56e-04		Exp: 4		Enob: 14.66
✓	N=32		SQNR: 90.00 dB		Max Err: 3.12e-04		Exp: 5		Enob: 14.66
✓	N=64		SQNR: 90.00 dB		Max Err: 6.25e-04		Exp: 6		Enob: 14.66
✓	N=128		SQNR: 90.00 dB		Max Err: 1.25e-03		Exp: 7		Enob: 14.66
✓	N=256		SQNR: 90.00 dB		Max Err: 2.50e-03		Exp: 8		Enob: 14.66
✓	N=512		SQNR: 90.00 dB		Max Err: 5.00e-03		Exp: 9		Enob: 14.66
✓	N=1024		SQNR: 90.00 dB		Max Err: 1.00e-02		Exp: 10		Enob: 14.66
✓	N=2048		SQNR: 90.00 dB		Max Err: 2.00e-02		Exp: 11		Enob: 14.66

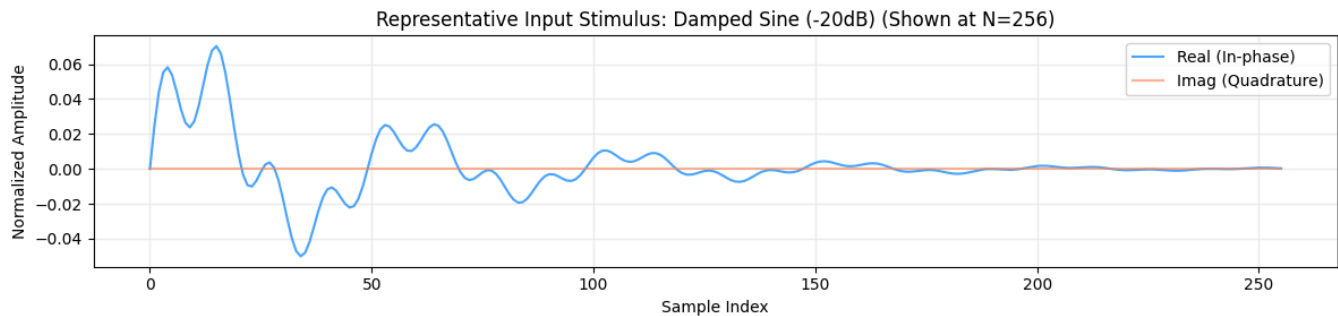
--- Scenario: Damped Sine (Full) ---



✓	N=16		SQNR: 70.36 dB		Max Err: 7.23e-04		Exp: 2		Enob: 11.40
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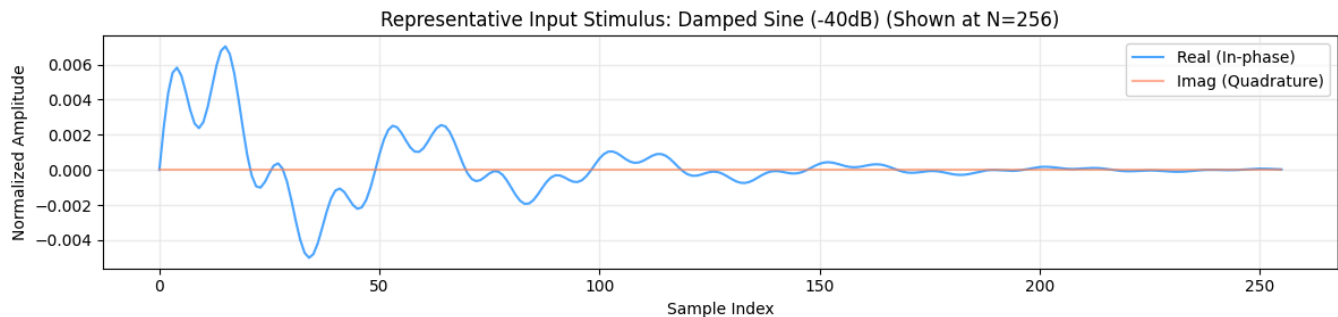
✓	N=32		SQNR: 68.52 dB		Max Err: 8.70e-04		Exp: 2		Enob: 11.09
✓	N=64		SQNR: 66.40 dB		Max Err: 1.34e-03		Exp: 3		Enob: 10.74
✓	N=128		SQNR: 64.96 dB		Max Err: 2.96e-03		Exp: 4		Enob: 10.50
✓	N=256		SQNR: 63.69 dB		Max Err: 6.94e-03		Exp: 5		Enob: 10.29
✓	N=512		SQNR: 61.92 dB		Max Err: 1.17e-02		Exp: 6		Enob: 9.99
✓	N=1024		SQNR: 60.25 dB		Max Err: 2.18e-02		Exp: 7		Enob: 9.72
✓	N=2048		SQNR: 58.13 dB		Max Err: 3.91e-02		Exp: 8		Enob: 9.36

--- Scenario: Damped Sine (-20dB) ---



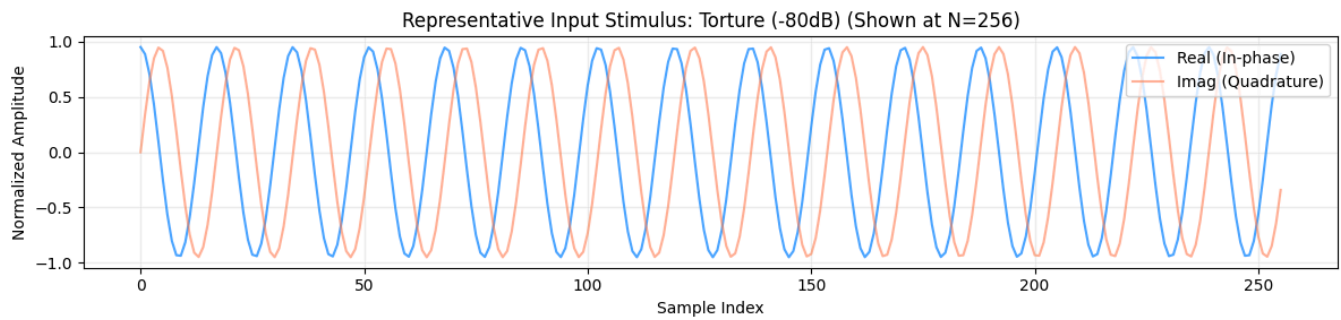
✓	N=16		SQNR: 61.85 dB		Max Err: 1.07e-04		Exp: 0		Enob: 9.98
✓	N=32		SQNR: 58.75 dB		Max Err: 2.22e-04		Exp: 0		Enob: 9.47
✓	N=64		SQNR: 58.23 dB		Max Err: 4.77e-04		Exp: 0		Enob: 9.38
✓	N=128		SQNR: 56.62 dB		Max Err: 9.14e-04		Exp: 1		Enob: 9.11
✓	N=256		SQNR: 56.14 dB		Max Err: 1.20e-03		Exp: 2		Enob: 9.03
✓	N=512		SQNR: 55.57 dB		Max Err: 1.62e-03		Exp: 3		Enob: 8.94
✓	N=1024		SQNR: 54.84 dB		Max Err: 3.08e-03		Exp: 4		Enob: 8.82
✓	N=2048		SQNR: 53.39 dB		Max Err: 4.96e-03		Exp: 5		Enob: 8.58

--- Scenario: Damped Sine (-40dB) ---



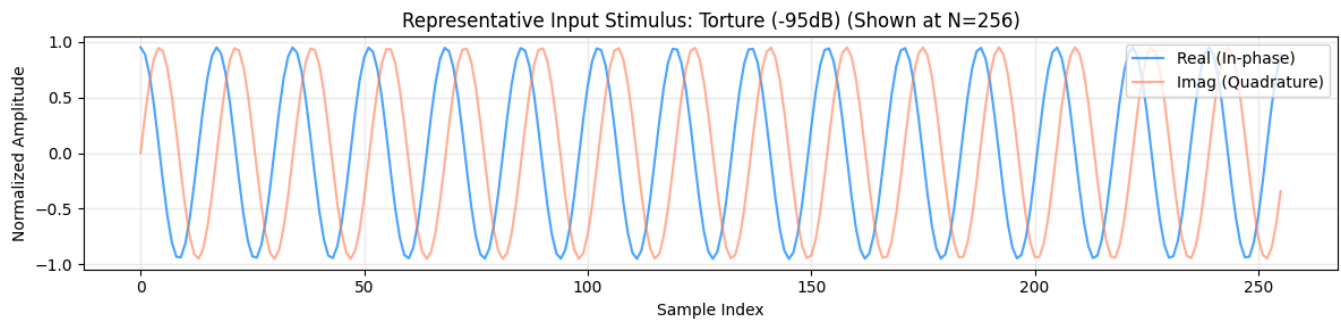
✓	N=16		SQNR: 39.61 dB		Max Err: 1.64e-04		Exp: 0		Enob: 6.29
✓	N=32		SQNR: 37.56 dB		Max Err: 2.95e-04		Exp: 0		Enob: 5.95
✓	N=64		SQNR: 37.38 dB		Max Err: 3.69e-04		Exp: 0		Enob: 5.92
✓	N=128		SQNR: 36.87 dB		Max Err: 6.91e-04		Exp: 0		Enob: 5.83
✓	N=256		SQNR: 36.97 dB		Max Err: 1.02e-03		Exp: 0		Enob: 5.85
✓	N=512		SQNR: 36.97 dB		Max Err: 1.53e-03		Exp: 0		Enob: 5.85
✓	N=1024		SQNR: 36.77 dB		Max Err: 2.36e-03		Exp: 0		Enob: 5.81
✓	N=2048		SQNR: 36.79 dB		Max Err: 3.34e-03		Exp: 1		Enob: 5.82

--- Scenario: Torture (-80dB) ---



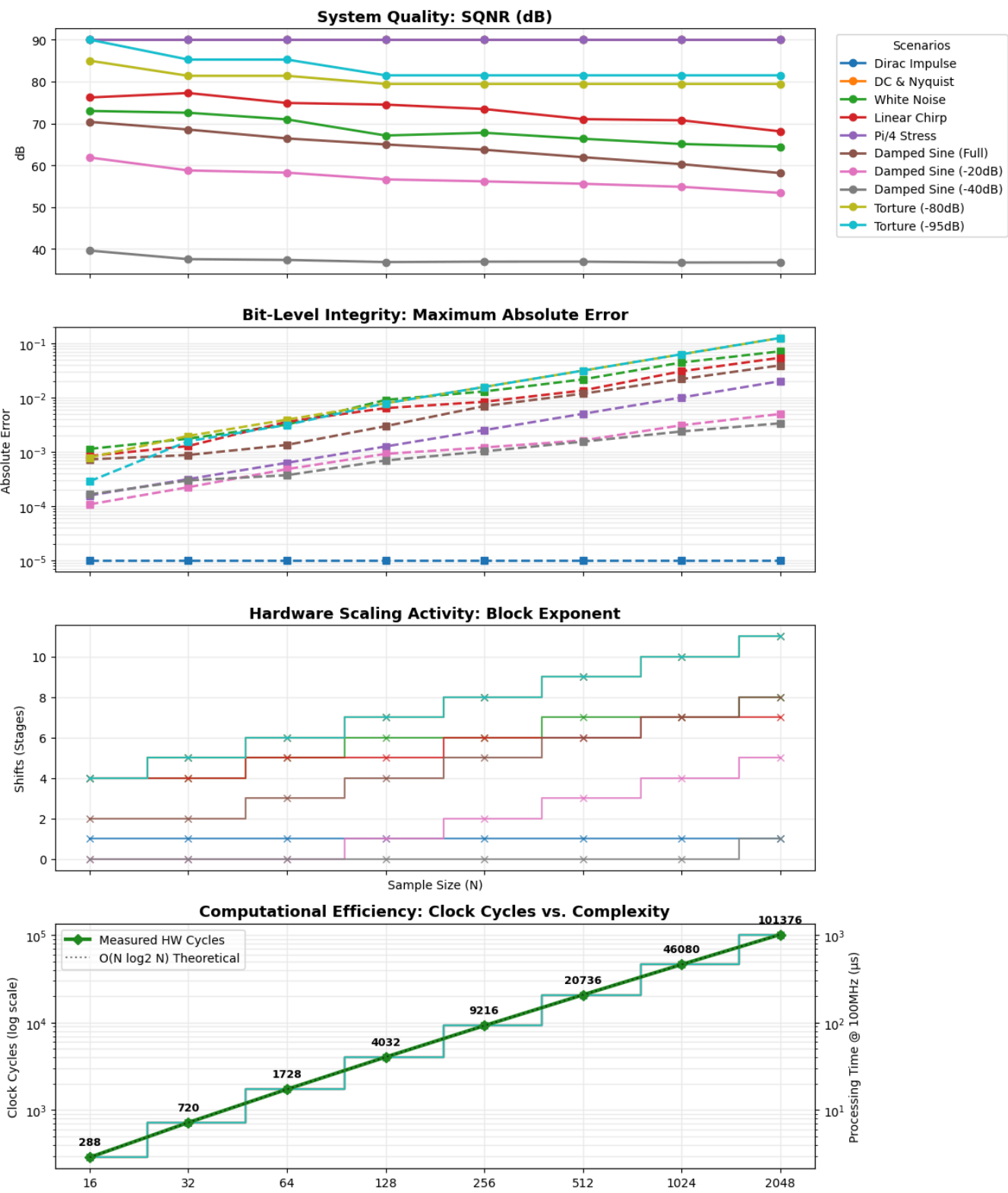
✓	N=16		SQNR: 84.97 dB		Max Err: 7.81e-04		Exp: 4		Enob: 13.82
✓	N=32		SQNR: 81.36 dB		Max Err: 1.95e-03		Exp: 5		Enob: 13.22
✓	N=64		SQNR: 81.36 dB		Max Err: 3.91e-03		Exp: 6		Enob: 13.22
✓	N=128		SQNR: 79.42 dB		Max Err: 7.81e-03		Exp: 7		Enob: 12.90
✓	N=256		SQNR: 79.42 dB		Max Err: 1.56e-02		Exp: 8		Enob: 12.90
✓	N=512		SQNR: 79.42 dB		Max Err: 3.12e-02		Exp: 9		Enob: 12.90
✓	N=1024		SQNR: 79.42 dB		Max Err: 6.25e-02		Exp: 10		Enob: 12.90
✓	N=2048		SQNR: 79.42 dB		Max Err: 1.25e-01		Exp: 11		Enob: 12.90

--- Scenario: Torture (-95dB) ---



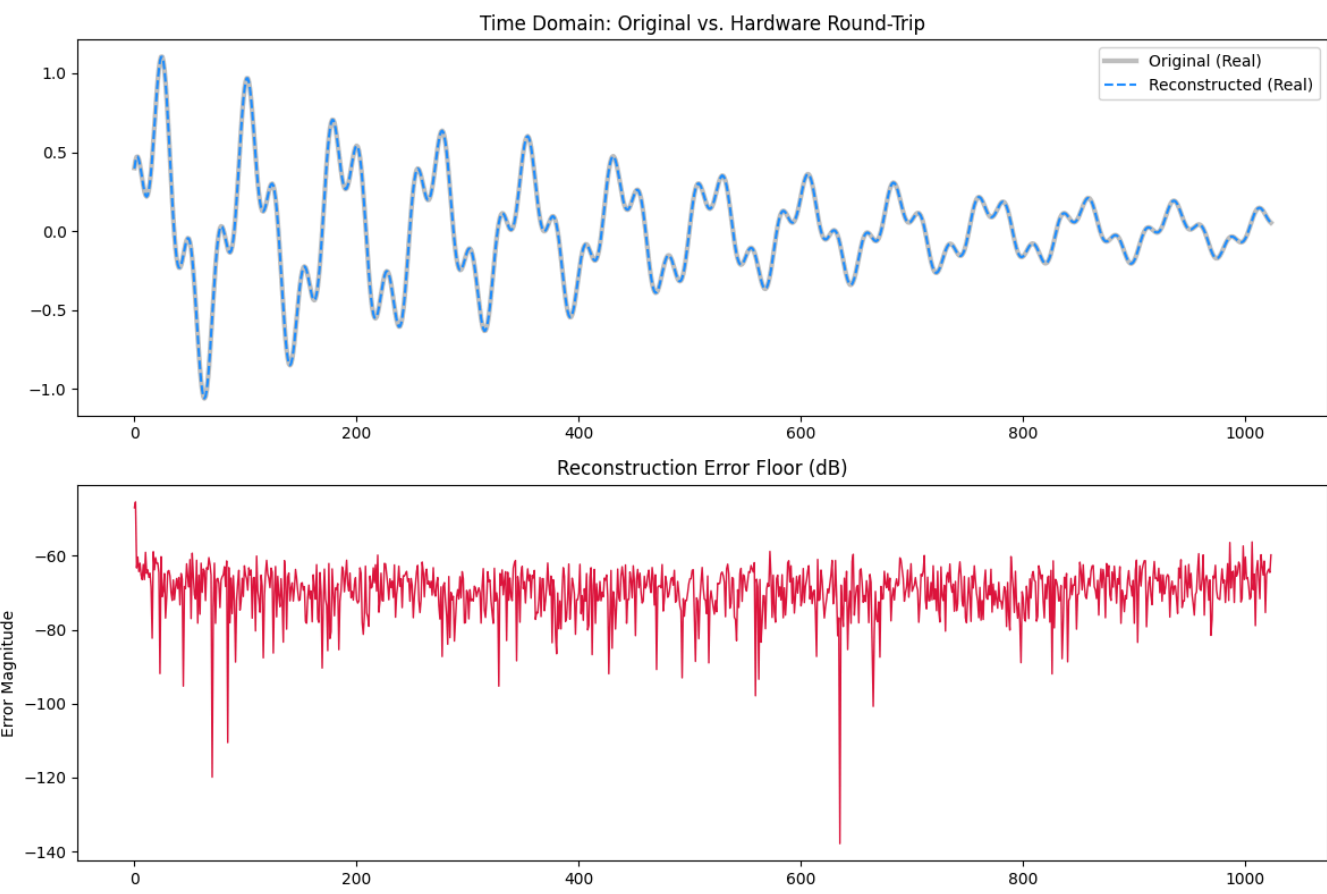
✓	N=16		SQNR: 90.00 dB		Max Err: 2.85e-04		Exp: 4		Enob: 14.66
✓	N=32		SQNR: 85.24 dB		Max Err: 1.56e-03		Exp: 5		Enob: 13.87
✓	N=64		SQNR: 85.24 dB		Max Err: 3.12e-03		Exp: 6		Enob: 13.87
✓	N=128		SQNR: 81.48 dB		Max Err: 7.81e-03		Exp: 7		Enob: 13.24
✓	N=256		SQNR: 81.48 dB		Max Err: 1.56e-02		Exp: 8		Enob: 13.24
✓	N=512		SQNR: 81.48 dB		Max Err: 3.12e-02		Exp: 9		Enob: 13.24
✓	N=1024		SQNR: 81.48 dB		Max Err: 6.25e-02		Exp: 10		Enob: 13.24
✓	N=2048		SQNR: 81.48 dB		Max Err: 1.25e-01		Exp: 11		Enob: 13.24

Multi-Metric Analysis Dashboard



System Roundtrip & Signal Integrity

✓ RECONSTRUCTION COMPLETE
Round-Trip SQNR: 55.72 dB
Effective Precision: 8.96 bits



Hardware Verification: Final Summary

Scenario	Objective	Max SQNR	Avg ENOB	Max Exp	Status
Dirac Impulse	Tests basic timing and unity gain; no twiddle math involved.	90.00 dB	14.66	1	✓ PASS
DC & Nyquist	Checks sign-extension and accumulation at frequency extremes.	90.00 dB	14.66	11	✓ PASS
White Noise	Stresses the complex multipliers with random, high-energy data.	72.97 dB	11.07	8	✓ PASS
Linear Chirp	Tests phase-factor (twiddle) accuracy across all bins.	77.25 dB	11.88	7	✓ PASS
Pi/4 Stress	Validates 45-degree multiplications where Sine and Cosine are equal.	90.00 dB	14.66	11	✓ PASS
Damped Sine (Full)	Real-world transient; tests scaling logic with high dynamic range.	70.36 dB	10.39	8	✓ PASS
Damped Sine (-20dB)	Mid-scale signal; checks precision with 3-4 bits of head-room.	61.85 dB	9.16	5	✓ PASS
Damped Sine (-40dB)	Small-scale signal; identifies the quantization noise floor.	39.61 dB	5.91	1	⚠ WARN (Noise Floor)
Torture (-80dB)	Tests Processing Gain; detecting a tiny tone near a large one.	84.97 dB	13.10	11	✓ PASS
Torture (-95dB)	Extreme sensitivity test; signal buried below the LSB.	90.00 dB	13.58	11	✓ PASS
SYSTEM ROUND-TRIP	End-to-end FFT -> IFFT verification; validates unity gain and total error.	55.72 dB	8.96	N/A	✓ PASS